

Project Concept Idea

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Project Title:	Predicting Customer Churn in Subscription Services Using Interpretable Machine Learning: A Business Decision-Support System

Project Summary

Customer churn, the voluntary discontinuation of a subscription or service, is a critical commercial challenge across the telecoms, SaaS, banking, and retail sectors. Research consistently demonstrates that retaining an existing customer costs 5–7 times less than acquiring a new one, yet most organisations lack predictive systems capable of identifying at-risk customers before they disengage. This project proposes the development of an interpretable machine learning pipeline that predicts customer churn probability from behavioural, transactional, and engagement data, and translates model outputs into actionable retention recommendations via a decision-support dashboard.

Three ensemble classifiers, Logistic Regression (baseline), Random Forest, and XGBoost, will be trained and compared. SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations) will be applied to produce per-customer churn driver explanations, making the system accessible to non-technical CRM and marketing stakeholders. The distinguishing academic contribution is the decision-support layer: the translation of ML predictions into prioritised, personalised retention intervention triggers, an integration not consistently present in existing churn prediction literature.

Literature Review

The following five peer-reviewed papers (2020–2024) contextualise the research and identify the gap this project addresses:

- [1] Vafeiadis et al. (2022). “A Comparison of Machine Learning Techniques for Customer Churn Prediction.” Simulation Modelling Practice and Theory / IEEE Access. Benchmark seven classifiers on telecoms data; XGBoost and Random Forest consistently outperform simpler models. This project extends their analysis with an interpretability layer absent from their work.
- [2] Lundberg & Lee (2017, widely cited 2020–2024). “A Unified Approach to Interpreting Model Predictions.” NeurIPS. Foundational SHAP methodology adopted in this project’s interpretability component.
- [3] Richhariya & Singh (2021). “A Survey of Various Churn Prediction Techniques.” IEEE International Conference on Computing, Communication, and Intelligent Systems. Reviews feature engineering approaches for churn; motivates this project’s multi-domain feature set design across telecoms and banking datasets.
- [4] Alboukaey et al. (2023). “Dynamic Behavior-Based Churn Prediction in the Telecommunication Industry.” Expert Systems with Applications. Introduces temporal behavioural features; directly informs this project’s feature engineering phase.
- [5] Devriendt et al. (2021). “A Literature Survey and Experimental Evaluation of the State-of-the-Art in Uplift Modelling.” Journal of Machine Learning Research. Introduces uplift modelling as an extension of churn prediction toward causal retention targeting, a stretch goal for this project’s decision-support layer.

Research Gap: While churn prediction accuracy has been extensively studied, the integration of XAI with a business-facing decision-support interface, translating SHAP outputs into prescriptive retention actions for non-technical stakeholders, is not present in the current literature.

Methodology

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Phase 1, Data Acquisition and Preprocessing:

Three publicly available, pre-labelled datasets will be used:

- IBM Telco Customer Churn Dataset (Kaggle), 7,043 customers; 20 features including tenure, contract type, monthly charges, service subscriptions. Widely used benchmark.
- Bank Customer Churn Dataset (Kaggle), 10,000 records; 13 features including credit score, balance, product count, and estimated salary. Cross-domain validation.
- E-commerce Customer Churn Dataset (Kaggle / UCI), 5,630 records; behavioural features including login frequency, last purchase recency, complaint history. Tests model generalisation across sectors.

Preprocessing: duplicate removal, one-hot encoding of categorical variables, Min-Max normalisation of continuous features, SMOTE for class imbalance (churn events are typically 15–30% of records), train/test split (80/20 stratified).

Phase 2, Feature Engineering:

Features will be organised across four categories:

- Behavioural: login frequency, product usage rate, feature adoption breadth, support ticket frequency
- Transactional: average monthly spend, payment method, billing cycle, outstanding balance
- Contractual: contract type, tenure, discount history, renewal history
- Engagement: NPS proxy scores (where available), complaint frequency, promotional response history

Phase 3, Model Training and Comparison:

- Logistic Regression: L2 regularisation, C parameter tuned via cross-validation. Interpretable baseline.
- Random Forest: 200 estimators, Gini criterion, max depth, and min samples tuned via Grid Search CV.
- XGBoost: learning rate, max depth, subsample, n_estimators tuned via Bayesian optimisation (Optuna).

All models evaluated using 10-fold stratified cross-validation. Final comparison on held-out 20% test set.

Phase 4, XAI Integration:

- SHAP TreeExplainer applied to Random Forest and XGBoost. Outputs: global feature importance bar chart, beeswarm summary plot, per-customer waterfall chart.
- LIME applied as secondary validation of SHAP feature attributions on a sample of 100 predictions.
- Churn driver categories derived from SHAP values (e.g., “High monthly charge”, “Short tenure”, “Month-to-month contract”) mapped to a retention intervention taxonomy.

Phase 5, Decision-Support Dashboard:

A Streamlit (or Flask + React) application will provide: customer churn probability score, top 3 SHAP-identified churn drivers, recommended retention intervention, and batch processing capability for CSV upload of customer lists. The interface will be designed for non-technical CRM users.

Evaluation Methods:

Quantitative: Accuracy, Precision, Recall, F1-score, AUC-ROC, MCC; Brier Score (probability calibration).

Any specific details and requirements

Datasets	IBM Telco Churn (Kaggle), Bank Customer Churn (Kaggle), E-commerce Churn Dataset (Kaggle/UCI), all freely available, pre-labelled, and well-documented
Python or R	scikit-learn, XGBoost, SHAP, LIME, Optuna, pandas, NumPy, Matplotlib, Seaborn, Streamlit or Flask (dashboard)
Web Framework	Streamlit (preferred, minimal frontend code required) or Flask + React for a more customised interface